

Kaylin(Huilin) Tang

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Education

2025.02—2027.02 **The University of Western Australia** **Perth, Australia**

Master of Information Technology | Specialisation: Applied Computing | Full-time

2017.09—2021.06 **Nanjing University of Aeronautics and Astronautics** **Nanjing, China**

Bachelor of Engineering | Major: Air Transportation Management | GPA: 3.8/5.0

Research Experience

2026.02—2026.06 **Civil Aviation Data Warehousing & Network Analytics** **Unit: Data Warehousing**

- Designed and implemented an end-to-end data engineering pipeline for large-scale civil aviation datasets, including ETL, dimensional modelling, and analytical data warehousing.
- Constructed a Star Schema and graph database architecture to support multidimensional analysis of flight operations, airport efficiency, and passenger behaviour.
- Modelled global airline networks in Neo4j and performed graph-based analyses using Cypher and APOC for route connectivity and transit analysis.
- Produced business insights through SQL analytics, Power BI visualisation, and association rule mining, demonstrating the application of data warehousing techniques to transportation systems.

2026.02—2026.06 **Machine Learning for Predictive Analytics** **Unit: Machine Learning**

- Implemented and evaluated classical machine learning algorithms for classification, regression, ensemble learning, and clustering using NumPy and scikit-learn.
- Built a Softmax Regression classifier from scratch with vectorised gradient descent and benchmarked its performance against the scikit-learn implementation on the MNIST handwritten digit dataset.
- Developed predictive models including Decision Trees, Random Forests, AdaBoost, Voting Classifiers, and Support Vector Machines, with systematic evaluation using accuracy, precision, recall, and F1-score.
- Applied feature importance analysis and unsupervised clustering to investigate model interpretability and data patterns in real-world healthcare and image datasets.

2020.03—2021.03 **Innovation Training Program for College Students** **Group Member**

- Publication: *Relationship between Evolution Characteristics of Aviation Carbon Emissions and Regional Economic Development*. *China Transportation Review*(2022).
- Applied the Logarithmic Mean Divisia Index (LMDI) decomposition method to quantify annual carbon emissions of major airports in the Greater Bay Area (2009–2016), based on data from the OAG aviation database.
- Analysed key drivers of aviation carbon emissions and proposed data-driven recommendations for route network optimisation, aviation technology improvement, and sustainable regional transportation development.
- This program won the First Prize of the 6th NUAA Undergraduate Academic Forum in December 2020.

Professional Experience

2026.06—2026.07 **University IT of UWA** **IT Intern**

- Contributed to the optimisation of the University's new student portal through user-centred analysis, usability evaluation, and evidence-based improvement recommendations.
- Developed practical experience in user-centred design, digital transformation, and enterprise software development within a large-scale university IT environment.

2021.07—2022.03 **Changzhou International Airport Group** **Marketing Assistant(Full-time)**

- Collaborated with online travel agencies (e.g., Trip.com and Ly.com) to plan and implement marketing campaigns supporting flight route promotion and passenger growth.
- Analysed airport shuttle operations and passenger-related data using Power BI to support operational monitoring and data-driven decision making.

Technical Skills and Additional Information

- Programming:** Python, R, C | **Databases:** MySQL, PostgreSQL, Neo4j
- Tools:** Power BI, Git, GitHub | **Computing:** Web Dev, Cloud Computing, High Performance Computing
- Languages:** English (IELTS 7.0), Japanese (Completed a short-term language program at Waseda University)
- Personal Interests:** Photography, Chinese Chess, Reading, Calligraphy